AgriSense: Research Synthesis, Selected Core Invention, and Roadmap

AgriSense should not build either report's full architecture. Its strongest defensible contribution is a narrower mechanism: an **evidence-graded, version-aware eligibility engine**. The engine works out whether a farmer is eligible, how strongly that is established, which conflicts block certainty, which single cheapest piece of evidence would settle it, and which past decisions must be recomputed when a GR or a land record changes. Zero-knowledge (ZK) attestation, spatial coalitions, bridge finance, satellite survival monitoring and carbon stacking should be either deferred as separate research tracks or simulated.

TL;DR

- **Selected core:** EVIDENT (Evidence-graded, Versioned Eligibility with Deterministic gap Targeting): deterministic rules-as-code compiled from Maharashtra GRs with LLM assistance and human approval; every input fact typed by provenance and assurance level (self-declared, OCR'd, signed-and-verified, registry-sourced); a multi-valued verdict, a minimum-cost "evidence gap" plan, and bitemporal impact analysis that re-evaluates only affected decisions when rules or evidence change. It is testable in an IPD prototype with only public GRs and synthetic profiles.
- **Why not the reports' headline ideas:** LLM-to-SHACL eligibility is already published (MDPI *Applied Sciences* 15(20):11235, 2025), and LLM-assisted rules-as-code compilers with policy-diff impact analysis already exist as open-source work. MSKVY 2.0 already specifies a collector-run GIS land-aggregation portal, and the coalition problem is a known NP-hard connected-subgraph variant. Anon Aadhaar proofs are real, but no evidence was found that any portal accepts ZK tokens, and the signature format of the digitally signed 7/12 could not be verified. ULI is a regulated-lender network, not a student-accessible API.
- **What to do next:** extend the existing AgriSense rule engine and OCR pipeline into EVIDENT first, starting with 15–25 MahaDBT schemes and their GR amendment histories. Measure false-positive eligibility rate, gap-plan efficiency and re-evaluation correctness against strong baselines (attribute filter, boolean rule engine, LLM/RAG-only, LLM→SHACL). Treat Anon Aadhaar predicates, co-holder consent gating and the coalition algorithm as scoped add-ons. Get IP advice before any public IPD exhibition.

Label key used throughout: **[R1]** = claim from Report 1; **[R2]** = claim from Report 2; **[V]** = verified externally in this research; **[I]** = my analytical inference; **[U]** = unverified or could not be confirmed.

Part I — Executive synthesis

Strongest conclusion. Plain scheme discovery is now commoditised. The central government launched Bharat-VISTAAR on 17 February 2026 in Jaipur; the launch was led

by Agriculture Minister Shivraj Singh Chouhan [V]. It is a voice- and chat-based AI platform covering scheme information, eligibility, application and grievance, and was launched in Hindi and English first [V]. It covers 10 central schemes: PM-KISAN, PMFBY, Soil Health Card, Modified Interest Subvention Scheme, SMAM, Per Drop More Crop, PMKSY, PM-AASHA, Agriculture Infrastructure Fund and KCC (Free Press Journal) [V]. Its FY2026-27 allocation of ₹150 crore was announced in the Union Budget 2026-27 and confirmed in a Rajya Sabha written reply on 13 March 2026 [V]. In Maharashtra, Chief Minister Devendra Fadnavis told the India AI Impact Summit on 20 February 2026 that MahaVISTAAR-AI "has already reached around 25 lakh farmers" (All India Radio) [V]. Two days later he was quoted saying it had reached 30 lakh (FintechBizNews), so the reach figure is self-reported and changing. AgriSense cannot win on "which schemes might apply to me".

Central opportunity [I]. What no surveyed system does well is say *how well* eligibility is established, *from which evidence, under which version of the rule*, and *what single action would most cheaply resolve the uncertainty*. Examples: a profile that says 2 ha while the uploaded 7/12 says 1.4 ha; a joint holding where only the farmer's share counts; a GR amendment that silently changes a threshold. These are exactly where farmers lose benefits or face rejection after a lottery selection. Report 1 identified this gap correctly. Report 2's critique that "DPI makes OCR discrepancy detection transitional" is partly right. However, AgriStack's own site says the registry is linked to land records "for non-legal, planning and advisory, and scheme-delivery purposes only" [V]. Farmers will therefore keep holding a mix of registry data, signed extracts, unsigned copies and self-declarations for years. An engine that reasons over *mixed-assurance* evidence stays useful as DPI matures, and gets better as more evidence moves to registry-sourced assurance.

Why this beats the alternatives.

- The verifiable-eligibility core is the only candidate whose central hypothesis can be tested with public data within an IPD timeline.
- Its novelty claim does not rest on a named technology. It rests on a specific interaction: assurance-typed evidence × multi-valued verdict semantics × minimum-cost gap targeting × bitemporal selective re-evaluation.
- Every other candidate depends on things that could not be verified or are unavailable: portal acceptance of ZK tokens, the 7/12 signature structure, ULI onboarding, lender participation, or sapling-level remote sensing.

Part II — Research inventory (both reports)

#	Proposal	Origin	Evidence level	Disposition
1	Pivot from recommendation to "verify, reconcile, prepare, manage lifecycle"	R1	Sound reasoning [I]	**Retained as framing**
2	LLM+RAG extraction of preconditions → SHACL/FOL → KG	R1	Published prior art: MDPI 15(20):11235 "A Framework for a Public Service Recommender System Based on Neuro-Symbolic AI" [V]	**Retained as a component, not as the novelty**; typed rule IR with human approval
3	Symbolic[Neuro] orchestration, explanation traces	R1	Standard design	**Retained**: LLM never in the runtime decision path
4	RASOR, Frontiers 2026, arXiv 2512.12109 claims (hallucination 75%→<40%)	R1	**[U]**: not verified	Do not cite until checked
5	Client-side ZK eligibility proofs (Groth16, Gnark/Arkworks)	R1	Anon Aadhaar proves feasibility for Aadhaar QR [V]	**Optional module** limited to Aadhaar-QR predicates
6	Dynamic zk-SNARKs (Dynalog, eprint 2026/144)	R1	**[U]**; unnecessary [I]	**Rejected**: expiry + regeneration is simpler and safer
7	Temporal DB with valid_from/valid_until, impact analysis on rule change	R1	Prior art exists (Catala date-scoped rules; RuleWeaver version-diff impact) [V]	**Retained, central**, combined with assurance typing
8	5-layer architecture incl. RBIH land-records API	R1	ULI land records are for onboarded lenders [V/I]	Retained conceptually; integrations simulated
9	VeriFarm paper and evaluation design	R1	Reasonable	**Retained and strengthened** (Part X)
10	DPI disintermediation critique; 3(k) critique	R2	Partly correct [V/I]	Accepted with qualification

#	Proposal	Origin	Evidence level	Disposition
11	Five Maharashtra bottlenecks table	R2	Mixed	See Part IV
12	Spatial-cadastral coalition (RAG + DBSCAN + BFS)	R2	NP-hard [V]; MSKVY 2.0 already has an aggregation portal design [V]	**Deferred as separate research track**
13	Fractional-share graph + multi-party digital NOC	R2	Co-holder consent requirement [V-secondary]	**Retained, narrowed:** share-aware eligibility + consent-gated state machine
14	zk-Satbara / zk-Aadhaar, 5 constraints incl. in-circuit transliteration and Poseidon nullifier	R2	Anon Aadhaar and zkPDF real [V]; 7/12 signature format **[U]**; in-circuit fuzzy transliteration impractical [I]	**Mostly rejected;** Aadhaar part optional
15	Purva-Sanmati bridge loan via ULI, vendor escrow	R2	No public third-party sandbox found [I]	**Simulation-only, deferred**
16	Sentinel-2 NDVI orchard-survival telemetry + auto Panchnama	R2	10 m pixels cannot resolve saplings [V]	**Deferred**; cannot substitute statutory verification frames doi
17	Stacked-benefit arbitrage incl. VM0042 carbon	R2	Issuance verified [V]; applicability and additionality doubtful [I]	**Rejected from core**; non-stacking checks kept in rule IR regrow
18	Independent Claim 1 (ZK + spatial + escrow)	R2	Bundles three inventions [I]	**Rejected**
19	Existing AgriSense features	User	Unknown completeness	**Retained; refactored** to feed EVIDENT

Boundary statement [I]. AgriSense now does profile capture, scheme ingestion, OCR, ranking and crop recommendation. The reports independently propose items 2–17. Combining compatible proposals enables provenance-typed evidence plus versioned rules plus multi-valued verdicts. Differentiation comes from minimum-cost gap targeting and selective bitemporal re-evaluation over assurance-typed evidence, with share-aware consent gating. Excluded from the core: dynamic SNARKs, blockchain registries, carbon arbitrage, live ULI bridge finance, and satellite-based survival adjudication.

Part III — Compatibility and contradiction analysis

Interaction	Classification	Reasoning
Neuro-symbolic extraction + temporal versioning	**Directly combinable**	Each extracted rule carries GR id, date, clause span and effective interval; LLM output is a *candidate* version executable only after human approval.
Evidence provenance + cryptographic attestation	**Needs intermediate mechanism**	A ZK proof shows a statement about data signed by a key, not that the data is current, legally valid, or about this parcel. Map proofs into the assurance lattice as one evidence type.
Privacy-preserving eligibility + spatial coalition	**Competing**	Coalition search needs neighbours' geometry, shares and crops, the opposite of data minimisation.
Spatial coalition + multi-party consent	**Complementary but independent**	Consent is needed for joint holdings anyway; coalitions multiply consenting parties and coordination cost.
Coalition feasibility + cross-scheme optimisation	**Keep separate**	Two layers of combinatorial uncertainty, no ground truth for either.
Pre-sanction finance + application state	**Conditionally compatible**	State machine can emit a "pre-sanction issued" event; finance stays a third-party decision.
Satellite telemetry + post-sanction fulfilment	**Conditionally compatible**	Block-level advisory warning only; never sets "survival met".
Policy changes + active applications and attestations	**Directly combinable (core)**	Dependency index maps attribute → rule version → decision → attestation; a change triggers recomputation and invalidation.
Privacy-preserving computation + auditability	**Tension**	Store salted hashes / commitments of evidence plus the decision trace; raw documents stay with the farmer.

Contradictions detected [I]:

1. **Report 2 contradicts itself on DPI.** It argues DPI makes document verification obsolete, then proposes zk-Satbara over PDF documents.

2. **ZK nullifiers conflict with joint holdings.** Hashing credential bytes + ULPIN + scheme id blocks legitimate co-holders from claiming their own shares of the same ULPIN, and fails across re-issued documents.
3. **"Dynamic ZK" conflicts with expiry semantics.** Anon Aadhaar's verifier already checks a signature-timestamp freshness window [V]; freshness is better handled by regeneration.
4. **The blockchain registry in EP3917076A1 [R1] is unnecessary.** A server-side spent-nullifier table is enough.

Part IV — Critical prior-art and feasibility findings

Verified facts and discrepancies

Topic	Report claim	What was found	Status
AgriStack Farmer ID	Digitally verifiable credential [R2]	agristack.gov.in: each farmer gets a FarmerID "and a digitally verifiable credential"; plot linkage is "for non-legal, planning and advisory, and scheme-delivery purposes only." agristack	**[V]**
AgriStack sandbox / UFSI	Open API gateway with sandbox [R2]	Sandbox gives simulated UFSI with sample data "to authorized stakeholders." UFSI API specs are publicly listed (e.g. `/agristack/verify-land`, get-farmer-id via Aadhaar hash). Farmer Registry FAQ: no external sharing without explicit consent given via e-sign. agristack + 2	**[V]**: specs public, access permissioned
Farmer ID counts	8.62 crore; target 11 crore [R2]	ICRISAT ISSCA: 6 crore Farmer IDs across 17 states as of 28 May 2025. 8.62 crore found only on an exam-prep site. icrisat c4scourses	6 crore **[V]**; 8.62 crore **[U]**
Maharashtra DCS	DCS replaces Girdawari [R2]	NeGD lists Maharashtra Digital Crop Survey under AgriStack. Farmers self-record via E-Peek Pahani; surveyors use DCS MH; Farmer Registry at mhfr.agristack.gov.in.	**[V]** negd agristack
Digitally signed 7/12	Signed with RSA/ECDSA, zkPDF-verifiable [R2]	Mahabhumi DSLR issues signed 7/12, 8A, Ferfar at ₹15, with a 16-digit verification number. A 23 Nov 2020 Revenue & Forest Dept circular (secondary sources) says QR + 16-digit-number signed 7/12/8A need no Talathi signature; a December 2025 Revenue Department announcement reaffirmed full legal validity. **Embedded PKCS#7/PAdES structure, signer and CA chain unverified.** DigiLocker issuance of 7/12 unconfirmed. ULPIN appears only where the parcel is geo-tagged (secondary); its length (11 vs 14 characters) is disputed. edgere + 4	Legal validity **[V-secondary]**; crypto format **[U]**

Topic	Report claim	What was found	Status
Anon Aadhaar	Circom/Groth16, mobile provers [R2]	PSE docs: circuit verifies the RSA signature of the Aadhaar Secure QR against UIDAI's key; ~30 s average browser proving; verifier SDK checks pubkey hash, a timestamp freshness window (example: < 1 hour), and a nullifier seed bound to the authorised frontend. pse + 2	**[V]**
zkPDF / DigiLocker	zkPDF and Anon DigiLocker exist [R2]	zkPDF (PSE) is a set of SP1 zkVM circuits checking PDF ByteRange digest and signature. PSE notes DigiLocker PDFs are RSA-SHA256 signed, while the QR uses symmetric encryption. hackmd hackmd "Anon DigiLocker" as a maintained library: **[U]**.	zkPDF **[V]**
MSKVY 2.0	₹50,000/acre/yr +3%; needs coalitions of small farmers [R2]	Scheme document (22 April 2023): rent "INR 1,25,000/Ha or 6% of ready reckoner rate whichever is higher with provision of 3% escalation annually" (≈₹50,600/acre). A "land aggregation portal shall be operated by the Nodal Agency through respective district collectors" mapping parcels within 5 km of substations; the Nodal Agency can sign back-to-back leases "with all landowners." Up to 5 MW at 33/11 kV within ~5 km; 5–25 MW at 132/33 kV within ~10 km. mahadiscom	Rent **[V]**; "individually ineligible" framing **wrong [I]**: each landowner leases own land; aggregation is institutional
Joint holdings	Own share only; all co-holders consent on ₹100 stamp paper [R2]	Marathi guides list "सामाईक क्षेत्र असल्यास सर्व खातेदारांचे संमतीपत्र" (consent of all co-holders, Appendix 19). For Fundkar: "संयुक्त मालकी असल्यास इतर खातेदारांच्या संमतीने शेतकऱ्यास स्वतःच्या हिश्याच्या मर्यादित लाभ घेता येईल" (benefit within own share, with co-holders' consent). marathipoint msdhulap ₹100 stamp paper not confirmed.	Share + consent **[V-secondary]**; stamp paper **[U]**
Pre-sanction workflow	100% upfront payment; strict windows [R2]	MahaDBT requires the पूर्वसंमती पत्र; purchases before pre-sanction get no subsidy. A secondary guide says drip/sprinkler documents are due within 10 days of	Partly **[V-secondary]**

Topic	Report claim	What was found	Status
		selection. mhsheti msdhulap Windows vary by scheme/GR.	
Fundkar Falbag	50/30/20; 90% irrigated / 80% dryland survival in yrs 2–3 [R2]	50:30:20 confirmed. **Survival rule conflicts across sources:** some say 80% in year 1 and 90% in year 2; others (incl. Marathi sources) say 90% irrigated / 80% dryland. Area 0.20–6 ha (non-Konkan), up to 10 ha in Konkan. bajajfinservmarkets + 2	**Conflict: verify against the GR**
ULI	89 lenders, 53 data providers, 141 data points [R2]	RBI's Trend and Progress Report (Business Standard, 29 Dec 2025): as of 12 Dec 2025, ULI had "64 lenders, including 41 banks and 23 NBFCs" (up from 36), using "over 136 data services… across 12 loan journeys", including "land records data from eight states". RBIH's later "What's New" page reports land records across 11 states, which reflects expansion after that date; it also lists NABARD e-KCC as integrated and all four CICs as live. 89/53/141 was found only on an exam-prep site. No third-party developer sandbox found.	Figures **conflict**; access **restricted [I]**
VM0042 India	First smallholder issuance, 57,463 credits [R2]	Regrow: 57,463 credits under VM0042 for Grow Indigo's Aadi project (VCS 2590), Punjab/Haryana, described as "the first smallholder soil carbon credit issuances under this standard globally." Business Standard (14 Sep 2026): Grow Indigo "paid out sums ranging between ₹3,000 and ₹15,000 to around 2,550 farmers" from its own funds because the credits were not yet fully sold; the credits accrued 2019–2022.	**[V]**; relevance to Maharashtra solar/orchard schemes **unsupported [I]**
MDPI 15(20):11235	Source for LLM→SHACL [R1]	Exists: LLM+RAG extraction of public-service preconditions into nested SHACL shapes validated against a citizen graph. doi mdpi	**[V]: direct prior art** mdpi

Topic	Report claim	What was found	Status
Kroll case	Kroll (2025) [R2]	Delhi HC, C.A.(COMM.IPD-PAT) 439/2022, 1 July 2025 (2025:DHC:5096): 3(k) refusal upheld. verdictum bananaip	**[V]**
CRI Guidelines 2025	Require technical effect [R1/R2]	Published by IPO; draft v2.0 26 June 2025. Test since *Ferid Allani* (Delhi HC, 12 Dec 2019): technical effect / contribution. devdiscourse mycrave	**[V]**
Coalition complexity	DBSCAN + BFS finds minimal coalition [R2]	MWCS is NP-hard and hard to approximate within a constant factor (El-Kebir & Klau); budget-constrained connected subgraph variants are NP-hard (Dilkina & Gomes, CPAIOR 2010).	**[V]**: heuristics must be framed as heuristics arxiv cornell
Sentinel-2 survival	NDVI predicts 90/80% survival [R2]	10 m resolution produces mixed pixels and is "challenging for small or clustered trees"; individual-tree mortality needs higher-resolution data.	**[V]**: saplings below pixel scale doi frames

Prior-art overlaps relevant to novelty [V/I]

- **LLM → formal eligibility rules:** MDPI 2025 (SHACL); NLLP 2025 workshop paper evaluating LLM translation of French legal text into Catala; RuleWeaver (2026 open-source) compiling sources via LLM into a typed IR with "type / reference / temporal / consistency checks" and "compare a revised policy version and show impacted rules/tests"; compiled-ai-labs tax-rules-compiler, which advertises "ambiguity refused." github + 3
- **Rules-as-code engines:** OpenFisca, used for benefit eligibility (e.g., Barcelona's "Les meves ajudes"); Catala, whose prioritized default logic handles exceptions and which found a real bug in OpenFisca. alphaxiv openfisca
- **ZK credentials over Indian IDs:** Anon Aadhaar; zkPDF-based DigiLocker verifiers. github zkmopro
- **Land aggregation:** MSKVY 2.0 land aggregation portal; MSAPL's GIS-based land bank portal (SKOCH case study); FPO-led KUSUM-A contiguous-plot aggregation in industry guides. google + 2
- **Satellite scheme verification:** PMFBY remote-sensing programmes exist; not verified in this research.

Implication [I]. "LLM extracts rules", "versioned rules", "ZK over Aadhaar" and "land aggregation" are each individually anticipated. Only their *specific interaction*, defined below, is a candidate contribution.

Part V — Alternative architectures and decision rationale

- **A — EVIDENT (verifiable, evidence-graded eligibility).** *Hypothesis:* typing evidence by assurance and computing multi-valued verdicts with minimum-cost gap plans lowers false-positive eligibility and evidence-collection burden versus boolean rule engines and LLM/RAG. Public GRs + synthetic profiles; no external dependencies. Prior-art exposure moderate (components known; interaction not found).
- **B — Spatial-cadastral coalitions + consent.** *Hypothesis:* a rooted, share-weighted connected-subgraph search finds feasible parcel blocks with fewer consenting parties than proximity clustering. Lawful cadastral-plus-share data is hard to get; MSKVY already aggregates institutionally; it frames individually eligible landowners as "ineligible" [I].
- **C — Fulfilment finance + environmental monitoring.** *Hypothesis:* bridge credit + NDVI warning reduces forfeiture. Needs ULI onboarding, lenders, ground-truth survival data and legal acceptance; Sentinel-2 cannot resolve saplings. researchgate
- **D — Report 2's integrated system.** B + C + zk-Satbara; depends on at least four unverifiable or unavailable services; the bundled claim invites unity and obviousness attacks [I].
- **E — Privacy-first ZK eligibility (Report 1's VeriFarm emphasis).** Feasible today only for Aadhaar QR; land predicates need a verified 7/12 signature format and a parcel-binding story; value is limited unless a verifier accepts the proof [I].

Criterion	A	B	C	D	E
Problem–solution strength	Strong	Weak–moderate	Real problem, weak solution	Diffuse	Moderate
Testability with public data	High	Low	Very low	Very low	Moderate (Aadhaar only)
Originality vs prior art	Interaction-level	Low–moderate	Low	Low (bundling)	Low
Privacy posture	Good	Poor (neighbour data)	Moderate	Mixed	Strong but narrow
IPD feasibility	High	Moderate (synthetic)	Simulation only	Low	Moderate

Selection: A as the core. E's Aadhaar predicate becomes one optional evidence type within A. B's share-and-consent logic is narrowed into A's state machine; B's coalition search is a separate optional study. C and D are deferred.

Part VI — Final unified invention: EVIDENT

Problem statement. A Maharashtra farmer's eligibility facts come from heterogeneous evidence of very different reliability: self-declaration; OCR of a photographed 7/12; a digitally signed 7/12 checked through Mahabhumi's 16-digit verification; an Aadhaar QR signature; consented registry data. The rules themselves change through GRs and amendments. Existing systems collapse all this into a boolean "eligible/not eligible" computed on whatever value they happen to hold. They do not say what is missing or conflicting, and they do not track which earlier decisions a rule change invalidates.

Central mechanism. Four parts:

1. **Assurance-typed evidence.** Every fact carries provenance (source, method, issuer, verification result), an assurance level from an ordered set, valid time and transaction time.
2. **Assurance-aware multi-valued evaluation.** Each rule atom declares the minimum assurance it needs. Evaluation returns one of {ELIGIBLE_ESTABLISHED, ELIGIBLE_PROVISIONAL, CONTINGENT, INDETERMINATE, CONFLICTED, INELIGIBLE_ESTABLISHED, INELIGIBLE_PROVISIONAL} with an explanation trace bound to GR clause spans.
3. **Minimum-cost gap targeting.** For a non-established verdict, compute the cheapest set of evidence acquisitions whose possible outcomes force a determinate verdict. Each acquisition has a cost: farmer effort, fees, travel.
4. **Bitemporal selective re-evaluation.** A dependency index (attribute → rule-version atoms → decisions → issued attestations) re-evaluates only affected decisions when a rule version, an evidence item, or an assurance (expiry, revocation, failed re-verification) changes. Every past decision can be replayed exactly.

Architecture (text diagram).

Actors and trust. The farmer is trusted for intent, not for facts; the device is trusted by the farmer; the backend is partially trusted and must minimise what it holds. Policy sources are authoritative but noisy, so changes are verified by hash and human review. UIDAI and Mahabhumi are trusted issuers only through verifiable channels. Co-holders are separate principals. MahaDBT is the authoritative decision-maker, and EVIDENT never claims sanction. Lenders are external and out of core.

End-to-end workflow.

1. **Policy ingestion:** GR fetched → hash compared → if changed, diff → LLM proposes IR changes with clause spans → reviewer approves → new RuleVersion with `valid_from`. Previous version closed at `valid_to`.
2. **Evidence acquisition:** farmer adds profile facts (A0), photographs documents (A1 via OCR), verifies a signed 7/12 by 16-digit number or QR (A2), scans Aadhaar QR (A2/A2-zk), or (simulated) pulls registry data with consent (A3).
3. **Evaluation:** for each candidate scheme, evaluate under the rule version valid on the evaluation date. Store decision = (verdict, trace, rule_version_ids, evidence_ids, tx_time).
4. **Gap planning:** if the verdict is not established, show the farmer one prioritised action, e.g. "Download the digitally signed 7/12 for Gat 104 (₹15) — it resolves both the area conflict and the joint-holding question."
5. **Consent gating:** if the land is joint, eligible area = own share. The application cannot reach READY_TO_SUBMIT until consent evidence from every listed co-holder is at the required assurance.
6. **Application preparation:** checklist, packet export, deadline tracking. Submission is by the farmer or CSC on MahaDBT.
7. **Revalidation:** events (rule version, evidence expiry, new 7/12 mutation/Ferfar, consent withdrawal) → dependency index → re-evaluate affected decisions → notify only on verdict change.

Inputs/outputs. *Inputs:* RuleVersions; EvidenceItems; cost table for acquisitions; evaluation date. *Outputs:* Decision records with verdict, trace, gap plan, state transitions, and change notifications.

Security boundaries. Raw Aadhaar, PAN and 7/12 images stay in the device vault by default. The backend stores derived attributes, a SHA-256 of each document, and the verification result. PAN is not needed for any scheme in scope, so do not collect it unless a specific rule requires it [I]. With the optional Anon Aadhaar proof, the backend learns only the revealed fields (e.g., state = Maharashtra, age ⩾ 18) plus a per-scheme nullifier.

Failure handling. Unknown is never converted to eligible.

- Unavailable source → evidence stays at its current assurance and the decision is marked stale-risk.
- Ambiguous policy clause → the rule atom is marked `INTERPRETATION_PENDING`, and any verdict depending on it is at most PROVISIONAL.
- Proof failure → the A2-zk evidence is discarded and the system falls back to the lower level.
- Withheld consent → CONTINGENT, with the blocker named.
- Failed external integration → no state transition.

Part VII — USP in three forms

1. **One sentence (IPD):** "AgriSense tells a farmer not just which schemes fit, but how solidly each eligibility is proven, what single document would settle any doubt, and automatically re-checks every past answer when a GR or land record changes."
2. **Technical USP:** a deterministic eligibility engine over bitemporal, assurance-typed evidence. It computes a seven-valued verdict with clause-level traces and a minimum-cost evidence-acquisition plan by exact search over the small atom set of each rule. It keeps an attribute→atom→decision dependency index, so that a rule-version or evidence change triggers re-evaluation only of affected decisions, with bit-exact replay of any historical decision. Joint-holding share arithmetic and co-holder consent are first-class evidence gates in the application state machine.
3. **Research positioning:** *Problem:* welfare-eligibility systems treat inputs as equally trustworthy and rules as static, which yields false-positive eligibility and silent staleness. *Gap:* rules-as-code and LLM-to-rules work (Catala, OpenFisca, LLM→SHACL) formalises rules, not evidence assurance or minimal resolution actions under policy evolution. *Contribution:* a formal evaluation semantics, a gap-planning algorithm and a selective re-evaluation index, instantiated on Maharashtra agricultural schemes with real GR amendment histories. *Evaluation:* false-positive eligibility rate, gap-plan cost, and re-evaluation correctness/latency against four baselines, with ablations.

Part VIII — Algorithmic specification

Definitions.

- Assurance levels, totally ordered: A0 self-declared < A1 extracted from an unverified document (OCR) < A2 issuer-verified document (signature/QR or verification-number checked; Anon Aadhaar proof counts here) < A3 authoritative registry response under consent.
- Evidence item e = (attr, value, unit, assurance, source, method, valid_from, valid_to, tx_time, doc_hash, expires_at).
- Rule version R = (scheme_id, version, valid interval, formula φ, provenance spans). φ is built from atoms a = (attr, op, threshold, min_assurance, tolerance) using AND, OR, NOT and prioritized exceptions (Catala-style: base rule overridden by exceptions in priority order).

Atom evaluation. At evaluation date t, take E_a = evidence for attr(a) valid at t and not expired.

1. If $E_a = \varnothing \rightarrow$ **U** (unknown).
2. Let m = max assurance in E_a, and E^* = items at level m.
3. If values in E^* disagree beyond tolerance → **C** (conflict). Also mark C if any lower-level item disagrees with E^* *and* m < min_assurance(a).

4. Otherwise let v = the consensus value and b = op(v, threshold). If m ⩾ min_assurance(a) → **T** or **F** (established); else → **T~** or **F~** (provisional).

Composition. A value set {T, T~, U, C, F~, F} plus CONTINGENT for atoms of type `future_event` (consent, pre-sanction, inspection).

- **AND:** F dominates; then C; then U; then F~; then T~; T only if all atoms are T. So a single provisional false yields F~ unless some atom is established-false.
- **OR:** the dual.
- **Exceptions:** an exception overrides only if its own guard evaluates to T. If the guard is T~/U/C, the result is downgraded to at most provisional or indeterminate.
- **Final mapping:** T→ELIGIBLE_ESTABLISHED; T~→ELIGIBLE_PROVISIONAL; F→INELIGIBLE_ESTABLISHED; F~→INELIGIBLE_PROVISIONAL; U→INDETERMINATE; C→CONFLICTED; any T-path blocked only by future_event atoms→CONTINGENT.
- **Invariant:** no path maps U or C to an eligible verdict.

Gap planning (formal). Let A_open = atoms whose value ∉ {T, F}. Each acquisition action k (e.g., "download signed 7/12", "scan Aadhaar QR", "obtain co-holder consent") has cost c_k and a coverage set cov(k) ⊆ attributes, which it can raise to assurance level lvl(k). Find the minimum-cost set S such that, after the actions in S, every atom on at least one decisive path has assurance ⩾ its minimum for *every* possible outcome value. Equivalently, the verdict becomes determinate regardless of what the documents reveal.

```
function GAP_PLAN(ϕ, evidence, actions):
    open = {a ∈ atoms(ϕ) : val(a) ∉ {T,F}}
    best = None
    for S in subsets(actions) ordered by cost(S):        # |actions|
typically ⩽ 12
        if cost(S) ⩾ cost(best): break
        upgraded = open where ∃k∈S: attr(a)∈cov(k) and lvl(k) ⩾
min_assurance(a)
        # worst-case determinacy: enumerate T/F outcomes for upgraded atoms
        if for all assignments σ ∈ {T,F}^upgraded:
                EVAL(ϕ, val with σ) ∈ {T, F, CONTINGENT}:
            best = S
    return best  (else return "cannot be resolved by available actions")
```

- **Complexity:** O(2^K · 2^M · |φ|), with K actions and M upgraded atoms. Per-scheme rules in MahaDBT-type schemes are small (estimate: M ⩽ 10, K ⩽ 12, to be measured), so exact search is cheap.
- **General case:** the problem is a weighted set-cover variant (NP-hard), so for larger rules use an ILP. Report the exact-vs-ILP crossover empirically.

- **Ranking variant:** rank actions by expected verdict-resolution per rupee, using a prior over outcomes from historical rates. This is a heuristic and must be labelled as such.

Bitemporal re-evaluation.

```
Index: attr → {(scheme, rule_version, atom)};  (scheme, rule_version) →
{decision_id}
on RuleVersionChange(scheme, old, new):
    Δ = atoms whose (attr, op, threshold, min_assurance) differ, plus
structural diff of ϕ
    affected = decisions(scheme, old) with valid time overlapping
new.valid_from
    for d in affected: if Δ touches an atom on d's decisive path or structure
changed:
        d' = EVALUATE(new ϕ, d.evidence_at(new.valid_from)); store d' (never
overwrite d)
        if verdict(d') ≠ verdict(d): notify; invalidate attestations bound
to d
on EvidenceChange(attr, farmer) / Expiry / Revocation:
    for (scheme, rv, atom) in Index[attr]: re-evaluate the farmer's live
decision
```

- **Correctness target:** the affected-set equals the set computed by full recomputation (recall = 1.0). Measure precision; unnecessary recomputations are harmless but costly.
- **Replay:** any decision can be reproduced from (rule_version_ids, evidence_ids as-of tx_time).

Share-aware land atom. eligible_area = Σ over the farmer's 7/12 entries of (share fraction × parcel area); share comes from the 7/12 holder entries, at the assurance of that 7/12 evidence. For a joint holding, a CONSENT(coholder_i) future_event atom is generated for each other listed holder. Unresolved share (names present, no shares recorded) → U, never an equal-split default; equal split may be shown only as a labelled what-if.

Optional module B' (research track): rooted share-weighted coalition search. Given adjacency graph G (polygons sharing a boundary within tolerance ε), root parcel r, node weight w(v) = area, node cost κ(v) = number of distinct consenting holders, and a feasibility predicate (distance to substation ⩽ D), find connected S ∋ r minimising Σκ subject to Σw ⩾ W. This is a rooted, quota-constrained connected-subgraph problem and is NP-hard (MWCS family). researchgate openproceedings Solve exactly via ILP with connectivity cuts for village-scale graphs (hundreds of nodes), with a greedy BFS heuristic as baseline. Do not describe DBSCAN+BFS as optimal.

Part IX — Prototype and implementation roadmap

Existing features: retain registration, myScheme ingestion (as a discovery feed), OCR (outputs become A1 evidence) and ranking (only within verdict classes). **Refactor** the eligibility engine into EVIDENT, and Aadhaar/PAN upload into device-vault + derived attributes (drop PAN unless required). Keep crop recommendation separate; it is not part of the contribution.

Stage 0 (2 weeks) — Corpus & ground truth. Collect GRs and MahaDBT scheme pages for 15–25 schemes, including at least 5 with documented amendments (candidates: Fundkar Falbag, mechanisation, drip/sprinkler, onion storage, shade-net). Two annotators encode gold rules; measure Cohen's κ. *Acceptance:* gold IR for every scheme, with clause spans.

Stage 1 (3–4 weeks) — Rule IR + compiler. Typed JSON/YAML IR with exceptions and validity intervals. LLM extraction → static checks (types, units, unreachable exceptions) → reviewer UI. Optional Catala/OpenFisca export. *Acceptance:* atom-level F1 against gold reported; no unapproved rule executable.

Stage 2 (3 weeks) — Evidence store + evaluator. PostgreSQL with bitemporal columns (valid_from/valid_to, tx_from/tx_to). Evaluator in Python with property-based tests (Hypothesis) for the invariant "U/C never yields eligible". *Acceptance:* 100% pass on invariant tests; deterministic replay.

Stage 3 (2 weeks) — Gap planner + consent gate + application state machine. States: DRAFT → EVIDENCE_PENDING → CONSENT_PENDING → READY_TO_SUBMIT → SUBMITTED (manual) → PRE_SANCTION_RECORDED → FULFILMENT_DOCS → CLOSED / REJECTED, with transitions guarded by verdict class. *Acceptance:* the planner returns the optimal plan on a hand-checked suite.

Stage 4 (2 weeks) — Change detector + dependency index. Replay real amendment histories as events. *Acceptance:* affected-set recall = 1.0 against full recomputation.

MVP demo milestone. Five synthetic farmers, including a joint holder, a profile/7-12 area conflict, and an expired document. Show verdicts, the one-action gap plan, a simulated co-holder consent, and a replayed GR amendment flipping one farmer's verdict with a notification.

Stage 5 (optional) — Anon Aadhaar evidence adapter. Use the published SDK with the test QR generator; record proving time on target phones. Map the proof to A2 for identity/state/age attributes, with a per-scheme nullifier seed. *Fallback:* skip; the core is unaffected.

Stage 6 (optional research track) — Coalition study on synthetic or openly licensed village polygons; ILP vs heuristics.

Deferred: ULI bridge finance (simulated webhook at most); Sentinel-2 monitoring (block-level advisory notebook at most); carbon stacking.

What the simulations do not prove [I]: simulated registry adapters do not prove UFSI access; the simulated consent gate does not prove legal validity of consent; no demo shows MahaDBT acceptance.

Technologies: FastAPI/Django backend; PostgreSQL (+PostGIS only if Stage 6); React/PWA client with offline queue; Tesseract or the existing OCR; OR-Tools / PuLP for ILP; @anon-aadhaar/core for Stage 5.

Part X — Scientific evaluation plan

H1 (false positives). EVIDENT's established-eligible verdicts have a lower false-positive eligibility rate than (B0) a myScheme-style attribute filter, (B1) a boolean deterministic rule engine on the same rules, (B2) LLM/RAG-only eligibility, and (B3) an LLM→SHACL pipeline in the style of MDPI 11235.

- *Dataset:* synthetic profiles (n ⩾ 1,000) with controlled injection of conflicts (profile vs document), missing documents, joint holdings and stale records.
- *Ground truth:* gold rules applied to the hidden "true" facts.
- *Metrics:* FP-eligibility rate; precision/recall per verdict class; coverage (share of cases with determinate verdict).
- *Statistics:* McNemar's test on paired decisions; bootstrap 95% CIs.
- *Negative result to watch:* EVIDENT may reduce coverage. Report the coverage–FP trade-off explicitly.

H2 (gap-plan efficiency). Gap plans require fewer acquisition actions and lower cost to reach a determinate verdict than (B4) the full static checklist and (B5) a greedy "ask for the most-missing attribute" policy. *Metrics:* mean actions, mean ₹/effort cost, fraction of cases resolved. *Ablation:* exact vs ILP vs greedy.

H3 (policy evolution). Selective re-evaluation achieves recall 1.0 on affected decisions, with lower compute than full recomputation, on replayed real GR amendment histories. *Metrics:* affected-set recall/precision, wall time, change-detection latency (GR publication → new RuleVersion approved; human-review time reported separately).

H4 (rule extraction). LLM-assisted compilation plus static checks reduces reviewer edits relative to LLM-only extraction. *Metrics:* atom-level F1; exception-structure accuracy; unsupported-claim rate (atoms without a source span); reviewer minutes per scheme.

Ablations: remove assurance typing (all evidence = A2); remove conflict state (latest value wins); remove temporal index (full recompute); remove share arithmetic (whole-holding area).

Confounders: synthetic distribution bias (document it and run sensitivity analysis over injection rates); annotator interpretation of ambiguous GR clauses (report as INTERPRETATION_PENDING counts).

Reproducibility: release the IR, gold rules, generator seed, code, and the GR corpus with hashes and dates.

Optional H5 (Stage 5): Anon Aadhaar proving latency and memory on 3 low-to-mid Android devices; proof payload size; server verification time. These are measurements, not claims.

Not claimed: any user-study outcome, adoption, or real-world rejection reduction. These require a field pilot with consent and ethics approval.

Part XI — Patent and publication strategy

Patent reality [V/I]. Section 3(k) excludes mathematical/business methods, computer programmes per se and algorithms. Courts since *Ferid Allani* (2019) and the CRI Guidelines 2025 require a technical effect or contribution beyond the normal running of a program. *Kroll* (Delhi HC, 1 July 2025) upheld refusal of a software system lacking such effect.
depenning + 2

Eligibility rules, checklists and benefit planning will very likely read as business methods/algorithms. The only arguable technical effects in EVIDENT are:

1. reduced recomputation and latency through the dependency-indexed selective re-evaluation, together with replayable bitemporal state;
2. reduced PII transmission and storage, through device-held raw documents plus issuer-signature verification into assurance-typed derived records.

Both are modest and close to prior art (incremental view maintenance, data minimisation patterns).

Recommendation [I]:

- Treat EVIDENT primarily as a **publication** contribution.
- Before any public IPD exhibition, poster, GitHub release or preprint, consult the DJSCE/SVKM IP cell about institutional ownership and inventorship (the policy was not verified here), and obtain a professional prior-art search under G06Q 50/26, G06Q 50/02, G06F 16/2477 (temporal data) and H04L 9/32.
- If counsel sees merit, file a **provisional** specification first, centred on the selective re-evaluation + assurance-typed evidence mechanism. Do not use Report 2's bundled claim.
- **Grace period:** Section 31 provides limited protection for certain disclosures (e.g., at notified exhibitions or papers read before learned societies) if an application follows within 12 months. Whether a college IPD exhibition qualifies was **not verified** here. Assume it does not, and file (or decide not to) before disclosure.

Paper directions.

1. **Main paper: "Evidence-Graded, Version-Aware Eligibility Determination for Agricultural Benefits: Semantics, Gap Planning, and Selective Re-evaluation."** *Gap:* rules-as-code ignores evidence assurance and policy-driven invalidation. *Method:* Parts VIII and X; Maharashtra corpus. *Related work:* Catala, OpenFisca, MDPI 11235, NLLP LLM→Catala, RuleWeaver-style compilers, benefits screeners. *Venues by fit:* ACM COMPASS; JURIX or ICAIL (legal informatics); NLLP workshop (extraction component); Digital Government: Research and Practice; Government Information Quarterly. *Required before submission:* gold corpus with agreement statistics; all four baselines; ablations.
2. **Focused algorithm/OR paper (optional): share-weighted rooted connected-subgraph search for parcel aggregation under consent costs.** Must engage the MSKVY 2.0 aggregation portal, the MWCS/budget-connected-subgraph literature (Dilkina & Gomes; El-Kebir & Klau) and land-consolidation research. *Venues:* Computers and Electronics in Agriculture; Land Use Policy; an OR-applications venue. *Required:* lawfully obtained or openly licensed cadastral data.
3. **Measurement/security note (optional): feasibility of ZK selective disclosure over Indian credentials for welfare eligibility.** Anon Aadhaar plus zkPDF on DigiLocker-signed PDFs, and the 7/12 signature structure once inspected. *Venues:* a security/privacy workshop or ACM COMPASS poster. *Required:* real device measurements and a threat model covering parcel binding and joint holdings.
4. **Remote sensing:** defer. A credible study would need ground-truth orchard survival counts and higher-resolution imagery.

Part XII — Final decision register

- **Selected core innovation:** EVIDENT. It combines assurance-typed bitemporal evidence, multi-valued verdict semantics with GR-clause traces, minimum-cost gap planning and dependency-indexed selective re-evaluation, plus share-aware consent gating.
- **Must retain:** the typed rule IR with human approval; the assurance lattice; the invariant "U/C never → eligible"; the gap planner; the bitemporal store and dependency index; share arithmetic and the consent gate; device-held raw documents.
- **Optional/deferred:** Anon Aadhaar evidence adapter; coalition search (separate study); simulated UFSI adapter; simulated pre-sanction finance webhook; block-level NDVI advisory.
- **Rejected:** dynamic zk-SNARKs; blockchain registries; in-circuit transliteration; ULPIN-keyed nullifiers; carbon/insurance arbitrage; satellite-determined survival entitlement; Report 2's bundled patent claim; "world-first" positioning.
- **Unresolved questions that could change the decision:**
    1. Does the signed 7/12 PDF carry an embedded, verifiable PKI signature? If yes, a zk-Satbara path becomes viable as a research track.

2. Will AgriStack UFSI sandbox access be granted to an academic team? If yes, A3 evidence becomes real.
3. What are the actual Fundkar survival thresholds in the current GR?
4. Do DJSCE/SVKM IP policies constrain publication?

- **Most important next experiment:** H1 on 15 schemes with injected conflicts, comparing EVIDENT against the boolean rule engine and LLM/RAG-only.
- **Strongest immediate prototype milestone:** a replayed real GR amendment that changes a threshold, with the dependency index flagging exactly the affected synthetic farmers and producing updated verdicts and one-action gap plans.

Part XIII — Evidence register (by origin)

Report 1 claims not independently confirmed: RASOR hallucination figures; Frontiers in AI 2026 and arXiv 2512.12109; Dynalog eprint 2026/144; EP3917076A1 and US20210297255A1 contents; DILRMP 3.0 ₹565.50 crore outlay.

Report 2 claims not independently confirmed: DAM outlay figures; 8.62 crore Farmer IDs; 30 crore plots by Kharif 2026; ULI 89/53/141; ₹100 stamp-paper consent; Magel Tyala ₹15,000 crore; CHC ₹10 lakh; Pack House group eligibility; exact MahaDBT upload windows.

Externally verified in this research: every fact marked [V] in Part IV. Sources: agristack.gov.in, the UFSI API listing and Farmer Registry FAQ; ICRISAT ISSCA; NeGD; the Mahabhumi DSLR help guide and 7/12 legal-validity circular coverage; the MSKVY 2.0 scheme document; PSE Anon Aadhaar and zkPDF documentation; MDPI 15(20):11235; the NLLP 2025 LLM→Catala study; Catala, OpenFisca and RuleWeaver; the Delhi HC Kroll judgment; IPO CRI Guidelines 2025; El-Kebir & Klau and Dilkina & Gomes; Sentinel-2 individual-tree literature; All India Radio and Free Press Journal (Bharat-VISTAAR, MahaVISTAAR-AI); Grow Indigo and Regrow; MediaNama and RBIH.

Original analytical inferences: all items marked [I], including the selection of EVIDENT, the contradiction analysis, the MSKVY reframing, the rejection decisions, and complexity bounds stated as estimates to be measured.

Caveats

- Several scheme facts rest on secondary Marathi guides rather than the governing GRs. Verify each against the GR before encoding it as gold truth.
- Report-cited 2026 preprints were not checked and should not be cited until read.
- No user study, field deployment or official integration has been done, and none is implied. All performance statements are hypotheses to be measured.
- The patent and disclosure comments are preliminary research observations, not legal advice.